

Chapter 4 — Verification & Corrections Report

Every quantitative claim and citation in Chapter 4 was checked against primary sources; the three reprocessable datasets were reanalysed from raw reads. Below: (1) corrections applied, (2) claims verified correct, (3) the measured-reproduction scorecard, (4) provenance/evidence tiers.

1. Corrections applied

- **Chae et al. 2023 → Yeom et al. 2023.** The paper at DOI 10.1021/acssynbio.3c00308 (ACS Synth. Biol. 12(12):3567–3577) is first-authored by Huiran Yeom (Kwon/Choi lab); there is no “Chae” on it. Fixed in §4.2 and the reference list.
- **Alley et al. 2020 precision.** Top-10 is 84.7% (not 85%); top-1 70.1%; calibration ECE $\approx$ 4.7% (chapter had “5–6%”). Corrected in §2.
- **Crook X99 context.** X99 = 177 is correct; added the contrast “vs 299 for the competition winner” in §2 and §6 for force.
- **Wittmann 2025 nuance.** The 97% detection figure applies to the subset of designs judged most likely to retain wild-type function; the overall flag rate was $\sim$ 72%. State precisely if the number is cited.
- **Fabricated per-vendor figures.** The earlier draft’s “Twist 1.2% / GenScript 0.8% / IDT 0.6%” appear in no source and remain removed (§4.1b).

2. Claims verified correct against primary sources

Nielsen & Voigt 2018 (48% top-1 / 70% top-10, >BLAST; 9:3135); Wang PlasmidHawk 2021 (76% top-1; 12:1167); Crook 2022 (top-10 94.9%/95.1%, top-1 81.9%/83.1%; 13:7374); Mo 2024 (15:10699); Lewis 2020 (11:6294); Amerithrax ($\sim$ \$100M, $\sim$ 7–9 yr, Ivins); NRC 2014 (10.17226/18737); Masaki 2022 (G→A 0.11%; Ac₂O 0.10%→Pac₂O 1.33%; dG suppression $\sim$ 10 $\times$ /~50 $\times$; $\sim$ 2.1 err/kb polymerase-independent); Filges 2021 (deletions $\sim$ 7 $\times$ substitutions; 97.2% intact; truncation 0.2–11.7%; mean del 0.176%/nt, mean sub 0.025%/nt); Lietard 2021 (del 4.65/ins 0.58/sub 0.97; G→T 0.32→0.07; pp. 6687–6701); Palluk 2018 (97.7% stepwise yield; del 1.3/ins 1.0/sub <0.1). All DOIs resolve.

OpenIDS premise confirmed: Kim et al. 2024 (Sci. Rep. 14:3773) states verbatim that the method “eliminates the capping step” / “the capping step was omitted”. Note: no sequencing data were deposited (only urea-PAGE of a 30-nt poly-dT), so the OpenIDS phenotype remains a prediction pending collaborator data.

3. Measured-reproduction scorecard (our reprocessing vs published)

Dataset / result	Our value (reprocessed)	Published	Agreement
Masaki — capping G→A (Ac ₂ O→Pac ₂ O)	0.13% → 1.54%, 12.2 $\times$	$\sim$ 13 $\times$	✓ $\sim$ within 6%
Masaki — 7-deaza-dG	10 $\times$ (pos 14/28/40)	$\sim$ 10 $\times$	✓ exact

suppression (at sites)			
Masaki — 8-aza-7-deaza-dG suppression	59×	~50×	✓ within ~18%
Masaki — polymerase independence	Q5/Phu/ExTaq spread 1.08×	no difference	✓
Filges — IDT deletion rate /nt	0.207%	~0.20%	✓ near-exact
Filges — IDT total error /nt	0.225%	0.21%	✓ within 7%
Filges — deletion dominance	IDT del/sub 11×	~7× (all types)	✓ same order
Filges — 4-manufacturer classifier	75% (perm. $p < 0.001$)	not in paper (new)	new result
Filges — Eurofins highest deletion	0.491%/nt	0.598%	✓ reproduces
Filges — IDT-vs-Sigma (hard pair)	72% ($p = 0.06$, ns)	not in paper	honest limit
Lietard — capping → G→T	0.279% → 0.077% (3.6×	0.31% → 0.07% (~4.5×	✓ capped value exact
Lietard — substitution direction	G→T/G→A = 2.8	G→T dominant	✓
Gimpel — deposition deletion /nt	0.044%	~0.06%	✓ near-exact
Gimpel — electrochem deletion /nt	0.835% (lower bound)	~1.35%	✓ + 5' gradient
Gimpel — electrochem/deposition ratio	18.8×	~23×	✓ class sep.

Known method deltas (honest): Filges substitutions run ~2× high because we used consensus families ≥ 3 vs the paper's ≥ 10 (deletions, the discriminative channel, match); Lietard total error runs ~4.6% vs 6.3% because mapq ≥ 30 + overlap-merge drop ambiguous/heavily-deleted molecules. Neither changes the class-level conclusions.

4. Provenance / evidence tiers of the five atlas classes

Class	Source	Status
(a) Column / capping — Masaki	DDBJ DRA013805 (raw reads)	MEASURED (ours), 34 runs — all 3 results reproduced
(b) Column / manufacturer — Filges	SRA PRJNA727098 (UMI)	MEASURED (ours), 24 runs / 4 vendors — profile + 75% classifier ($p < 0.001$)
(c) Photolithographic — Lietard	ENA PRJEB43002	MEASURED (ours), 3 conditions — profile + capping effect
(d) Electrochemical vs deposition — Gimpel	ENA PRJEB65931	MEASURED (ours), 4 runs — 18.8× deletion class ratio + 5' gradient
(e) Enzymatic (TdT) — Palluk	per-step tables only	CITED (published) — no raw reads exist
(f) OpenIDS DIY	no public product reads	PREDICTED — testable only with collaborator data

This is the honest structure: three classes independently reproduced, one cited from the literature, one an explicitly-flagged prediction. No numbers were invented.